

Homework 2

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MATH 421 – The Theory of Single Variable Calculus
Section 003

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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used.

Problem 1

Determine whether the following statements are true or false and provide a short justification.

A. $\{\{\emptyset\}\} \cup \emptyset = \{\emptyset, \{\emptyset\}\}$

Solution: The left-hand side is $\{\{\emptyset\}\} \cup \emptyset$. Since the union with the empty set adds nothing, this is just $\{\{\emptyset\}\}$. This set has exactly *one element*, namely the set $\{\emptyset\}$.

The right-hand side is $\{\emptyset, \{\emptyset\}\}$, which has *two elements*: $\emptyset$ and $\{\emptyset\}$.

Since the left-hand side has one element and the right-hand side has two, the sets are not equal.

Therefore, the statement is **false**.

B. $\{\{\emptyset\}\} \cup \{\emptyset\} = \{\emptyset, \{\emptyset\}\}$

Solution. The left-hand side is the set $\{\{\emptyset\}, \emptyset\}$, which has two elements: $\emptyset$ and $\{\emptyset\}$. The right-hand side is also $\{\emptyset, \{\emptyset\}\}$. Since both sides contain exactly the same two elements, the equality holds. **True**.

C. $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\emptyset\}$

Solution. The first set has elements $\emptyset$ and $\{\emptyset\}$; the second has $\{\emptyset\}$ and $\{\{\emptyset\}\}$. The only common element is $\{\emptyset\}$, so

$$\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\{\emptyset\}\}.$$

This is not $\{\emptyset\}$. **False**.

D. $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\{\emptyset\}\}$

Solution. As computed above, the intersection consists of the single common element $\{\emptyset\}$. Hence the intersection is $\{\{\emptyset\}\}$. **True.**

Problem 2 (Ross 1.11)

For each $n \in \mathbb{N}$, let P_n be the assertion $n^2 + 5n + 1$ is an even integer.

(a) **Show P_{n+1} is true whenever P_n is true.**

Proof. We compute

$$\begin{aligned} (n+1)^2 + 5(n+1) + 1 &= n^2 + 2n + 1 + 5n + 5 + 1 \\ &= n^2 + 7n + 7 \\ &= \underbrace{(n^2 + 5n + 1)}_{\text{the } P_n \text{ expression}} + \underbrace{(2n + 6)}_{= 2(n+3) \text{ even}}. \end{aligned}$$

If P_n holds, then $n^2 + 5n + 1$ is even; adding the even number $2(n+3)$ yields an even number. Hence P_{n+1} holds whenever P_n holds. $\square$

(b) **For which n is P_n actually true? What is the moral?**

For any $n \in \mathbb{N}$, whether n is even or odd, the value $n^2 + 5n + 1$ is odd:

$$\begin{cases} n \text{ even} \Rightarrow n^2, 5n \text{ even} \Rightarrow n^2 + 5n + 1 \text{ odd,} \\ n \text{ odd} \Rightarrow n^2, 5n \text{ odd} \Rightarrow n^2 + 5n \text{ even} \Rightarrow n^2 + 5n + 1 \text{ odd.} \end{cases}$$

Thus P_n is *false for every* $n \in \mathbb{N}$.

Moral. The inductive implication $P_n \Rightarrow P_{n+1}$ can be perfectly correct even if no P_n is true. Induction requires a true base case *and* the inductive step; the step alone proves nothing.

Problem 3

Prove using induction: If $x \in \mathbb{R}$ and $x \neq 1$, then

$$\sum_{i=0}^n x^i = \frac{1 - x^{n+1}}{1 - x} \quad \text{for every } n \in \mathbb{N}.$$

Proof. We proceed by induction on n .

Base case $n = 0$. We have

$$\sum_{i=0}^0 x^i = x^0 = 1 \quad \text{and} \quad \frac{1 - x^{0+1}}{1 - x} = \frac{1 - x}{1 - x} = 1$$

(since $x \neq 1$). Hence the claim holds for $n = 0$.

Inductive step. Assume for some $n \geq 0$ that

$$\sum_{i=0}^n x^i = \frac{1 - x^{n+1}}{1 - x}.$$

Then

$$\sum_{i=0}^{n+1} x^i = \left(\sum_{i=0}^n x^i \right) + x^{n+1} = \frac{1 - x^{n+1}}{1 - x} + x^{n+1} = \frac{1 - x^{n+1} + x^{n+1}(1 - x)}{1 - x} = \frac{1 - x^{n+2}}{1 - x}.$$

Thus the formula holds for $n + 1$, and by induction it holds for all $n \in \mathbb{N}$. $\square$

Problem 4 (Ross 1.9)

(a) For which integers is $2^n > n^2$ true?

Small cases. For $n \leq -1$, $2^n \in (0, 1)$ while $n^2 \geq 1$, so $2^n \not> n^2$ (false). For $n = 0$, $1 > 0$ (true). For $n = 1$, $2 > 1$ (true). For $n = 2, 3, 4$ we have $4 \not> 4$, $8 \not> 9$, $16 \not> 16$ (all false). For $n = 5$, $32 > 25$ (true). Part (b) shows the inequality is then true for every larger n .

Hence, for all integers,

$$2^n > n^2 \iff n \in \{0, 1\} \cup \{n \in \mathbb{Z} : n \geq 5\}.$$

(If restricting to positive integers, the answer is $n = 1$ or $n \geq 5$.)

(b) **Induction proof for $n \geq 5$.** We prove by induction on n that $2^n > n^2$ for every $n \geq 5$.

Base case $n = 5$. $2^5 = 32 > 25 = 5^2$.

Inductive step. Fix $n \geq 5$ and assume $2^n > n^2$. Then

$$2^{n+1} = 2 \cdot 2^n > 2n^2.$$

Since

$$2n^2 - (n+1)^2 = n^2 - 2n - 1 = (n-1)^2 - 2 \geq 4^2 - 2 = 14 > 0,$$

we have $2n^2 > (n+1)^2$. Therefore $2^{n+1} > (n+1)^2$.

By induction, $2^n > n^2$ holds for all $n \geq 5$. Combined with the small-case analysis in part (a), this fully answers the question.

Moral. Induction can be started at a convenient threshold n_0 ; once a statement is true at n_0 and implies truth at $n_0 + 1$, it propagates to all $n \geq n_0$ even if it fails for earlier n .

Problem 5

Theorem 1. Suppose A and B are subsets of a set S . If $A \subseteq B$, then $B^c \subseteq A^c$, where complements are taken relative to S (i.e., $A^c = S \setminus A$).

Proof. Let $x \in B^c$. By definition, $x \in S$ and $x \notin B$. If, towards a contradiction, we had $x \in A$, then since $A \subseteq B$ we would get $x \in B$, contradicting $x \notin B$. Hence $x \notin A$, which means $x \in A^c$. Therefore $B^c \subseteq A^c$. $\square$

Problem 6

Prove, using a proof by contradiction, that

$$\{2k + 1 : k \in \mathbb{N}\} \cap \{4k : k \in \mathbb{N}\} = \emptyset.$$

Proof. Suppose, for contradiction, that the intersection is nonempty. Then there exists an integer n and integers $a, b \in \mathbb{N}$ such that

$$n = 2a + 1 \quad \text{and} \quad n = 4b.$$

From $n = 4b$ we have $n = 2(2b)$, so n is even. But from $n = 2a + 1$ we have that n is odd. This is impossible, since no integer is both even and odd. Therefore our assumption was false, and the intersection is empty:

$$\{2k + 1 : k \in \mathbb{N}\} \cap \{4k : k \in \mathbb{N}\} = \emptyset.$$